

Md. Rasul Islam Bapary

Artificial Intelligence Engineer

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Work Experience

Softworks Limited, Dhaka | AI Engineer

May 2025 - present

- Designed and Developed a **VisionPOS** pipeline for **medicine image recognition** using **deep feature embeddings** and **OCR**. Extracted embeddings for indexing and similarity search to retrieve medicine metadata from a database. Integrated OCR for text-based retrieval to verify image embedding output outlining a combined vision system to recognize medicine.
- Developed a **smart card recognition system** to verify existing users and register new users. Implemented front and back card scanning to extract and validate selected information using **Bangla and English OCR, MRZ scanning, and 2D barcode** reading technologies. [\[Live\]](#)
- Developed a **smart referral web application** to streamline patient referrals from Upazila Health Complexes across Bangladesh to tertiary hospitals nationwide. Implemented **voice-based** referral input using **Speech-to-Text (STT)** models to **extract meaningful clinical keywords** from spoken data and **automatically populate government medical referral forms**. All AI models were **locally deployed** to ensure data security and system reliability. [\[Live\]](#)
- Developed **PharmaOrder**, a web application for managing medicine orders between wholesalers and retailers. Enabled retailers to place **urgent medicine requests** and allowed wholesalers to review, modify, and approve orders through a streamlined workflow. Ensured secure and efficient system performance with **locally deployed AI models**. [\[Live\]](#)
- Developed a **full-stack** voice-driven data extraction platform using **React** and **FastAPI**, supporting **Bengali speech-to-English** processing. Integrated **ASR**, LLM-based structuring (**Ollama**), domain-specific validation, and SQLite storage. Designed a scalable, **multi-workflow architecture** adaptable for **healthcare referrals, order management**, and other configurable use cases. [\[Live\]](#)
- Developed an automated **vehicle tracking system** for apartments in Dhaka, enabling license plate detection, recognition, and recording of vehicle information.

Bangladesh Machine Learning Institute, Dhaka | ML Engineer

May 2024 - April 2025

- Designed and developed an **Automatic Quality Control System** pipeline for garments, creating detection models for broken stitches, open seams, dirt marks, and foreign objects.
- Developed an API to calculate garment measurements from images, efficiently measuring waist, hem, front rise, back rise, and inseam on denim pants.
- Developed a **Trainer Agent** capable of training and fine-tuning LLMs with **DPO** and **GRPO** dataset and an **Evaluator Agent** to assess model performance using various metrics.

NBSoft, Rangpur | ML Engineer Intern

Mar 2023 - Nov 2023

- Developed Real-time **Crane and Container detection** model from camera footage.
- Built **Pacemaker detection** model from human chest x-ray.
- Implemented **Segmentation of Bones** model from pneumonia patients x-ray image.
- Researched and Built **Local Attention Grouping model** for **Gastritis Atrophy** and **Intestinal Metaplasia** classification from **Endoscopy Imaging**.

Skills and Expertise

- Programming Languages:** C, C++, Java, Python, HTML5, CSS3, JavaScript, Bash/Shell scripting.
- Frontend & Web Development:** React.js, Vite, Next.js, Material-UI.
- Frameworks & Libraries:** PyTorch, TensorFlow, Keras, OpenCV, PaddlePaddle, Scikit-learn, Flask, FastAPI, Tkinter, Streamlit.
- AI, ML & Specialized Technologies:** Computer Vision, Machine Learning, Large Language Models (LLMs), Generative AI, Reinforcement Learning, LangChain, RAG, OCR, STT, ASR, Data Extraction, HF Transformers, Prompt Engineering, NLP tasks.

- **Cloud & Compute Platforms:** Google Cloud, Microsoft Azure, Mila Cluster, Lightning AI, Anyscale, CI/CD (GitHub Actions).
 - **Databases & Data Management:** MySQL, PostgreSQL, SQLite3.
 - **DevOps & Tools:** Docker, Git, Amazon S3, VS Code, Cursor, Antigravity, Postman, PyCharm.
 - **Embedded Systems & IoT:** Raspberry Pi, Arduino, Sensors.
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Education

Begum Rokeya University, Rangpur
B.Sc. in Computer Science & Engineering
Obtained CGPA: 3.39 (**Rank: 7th**)

Rangpur-5404, Bangladesh
Year of Graduation: 2023

Publication

- Hasan, K.; Tuhin, K.A.; **Bapary, M.R.I.**; Doula, M.S.U.; Alam, M.A.; Ahad, M.A.R.; Uddin, M.Z. MMF-Gait: A Multi-Model Fusion-Enhanced Gait Recognition Framework Integrating Convolutional and Attention Networks. Symmetry 2025, 17, 1155. <https://doi.org/10.3390/sym17071155>
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Research & Project

- Developed a **AI Finance Accountant Agent** that uses **voice recognition** to query and **retrieve key data** from bank and company **financial statements**, streamlining information access and **asset** management. [\[Live Link\]](#)
 - **Customizable chatbot** with **Ollama** models. [\[SourceCode\]](#)
 - **Text Summarization** from **CNN Daily News** dataset. [\[SourceCode\]](#)
 - **Sentiment Analysis:** Coursera Experiments. [\[SourceCode\]](#)
 - **Hand-Raised person detection and identification** from video imaging. [\[SourceCode\]](#)
 - Real-time human **2D and 3D pose and 3D mesh** generation from webcam video. [\[SourceCode\]](#)
 - Bolt Nut **Recognition** with **center measurement**.
 - Classification of Corn Leaf Disease. [\[SourceCode\]](#)
 - Classification of **Autistic** Children. [\[SourceCode\]](#)
 - **Koala Detection** from the Wild. [\[SourceCode\]](#)
 - Fake Chest **X-ray Image Generation** using **VAE** and **Stable Diffusion**. [\[SourceCode\]](#)
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References

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